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Popular large language models like Qwen or LLaMA follow the already well-established and widely adopted paradigm of autoregressive decoding. This, however, has its inherent flaws: scaling poorly, needing to be retrained for specific sentiment control, and not being able to generalize knowledge they learned to be able to reverse a thought. Diffusion-based models aim to solve a lot of those issues. Although they are not as popular yet, research strongly suggests that they can be scaled to the size of and used in combination with autoregressive models, making them perform on par and sometimes better than their pure autoregressive counterparts. Models like LLaDA, Dream, and TiDAR have different approaches, but each offers advantages over autoregressive models, paving the way for future development of diffusion language models.

The landscape of language processing has long been dominated by autoregressive large language models such as Qwen [14], LLaMA [4] and the GPT series. These models process and generate language in a well ordered manner, starting with a token and continuing with the next one. This approach shows reliability but scales linearly for long sequence generation, as the model must predict token by token in each iteration [7]. In contrast to language, other applications, such as image generation, use a diffusion process. This different process generates multiple, in the case of images, pixels per iteration at once instead of generating pixels in sequence. Furthermore, diffusion models inherently support controllable generation via classifiers to control the look or subject of the image. Autoregressive models, however, lack this flexibility, needing to be expensively trained to change their toxicity or sentiment [12]. Consequently, there has been more research into using the diffusion process in language processing instead of, or in combination with, autoregressive models. This research, comparing diffusion language models to autoregressors, demonstrated that standard diffusion models following architectures such as masked diffusion offer parallelism and bi-directionality while also excelling in planning and control. Even though they offer near-constant inference time for an arbitrary sequence length, this constant speed results in them struggling with coherency and correctness [8]. This has led to hybrid approaches, which use a combination of both diffusion and autoregressive models, such as the model mentioned in Diffusion Guided Language Modeling

There are many types of diffusion language models that use different ways of computing data. In this report, discrete diffusion models will be discussed, in particular the class of masked diffusion models (MDMs). This class of models operates on a sequence of masked tokens, which then get iteratively unmasked. In practice that means, at first, all tokens in a sequence have the possibility to be everything, and the MDM then decides for arbitrary tokens in the sequence what they should be exactly. Another class of models is continuous diffusion models that were popular in early research but got replaced by discrete models rather quickly [7]. Discrete diffusion models were then researched and optimized; having breakthroughs like with SEDD [10] that allowed them to perform on par with autoregressive models. MDMs in particular demonstrate being performant and scalable [13] [19]. Furthermore, the combination of both autoregressive models and diffusion models, as suggested by Lovelace et al. [11], Lovelace et al. [12] and Liu et al. [9] will be discussed as well, as they show a good use of the strengths of both paradigms. Li et al. [7] gives a good overview of the current landscape of diffusion language models, providing essentially an exhaustive list of DLMs together with their specialties. This report won't cover all of them, although it is worth mentioning that they also cover models using different inference techniques as well as multimodality in depth. The most important factors that play into the comparison of the different approaches are summarized in Table 1 which shows all three methods that will be compared in this report based on certain important criteria.

To understand the main theoretical advantage of diffusion language models (DLMs) over autoregressive language models (ARMs), one has to look at how both paradigms work internally. ARMs generate tokens in sequence, like writing them out on a piece of paper. This mirrors the way we humans communicate language. However, it is inefficient when it comes to longer text sequences, as we first have to generate a token x_1 before generating a token x_2 , as the model has to base each token on the ones before it. Diffusion models aim to solve this. This paradigm of data generation stems from the image generation domain and means that for a sequence of tokens x_n , we generate multiple tokens in parallel. In theory, this improves the inference from linear time to constant

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	Autoregressive Models	Pure Diffusion Models	Hybrid Models
Modelle	LLaMA 3, Qwen3	LLaDA, Dream	SStar-LDM, TiDAR
Generation	Serial (Token-by-token): $x_i \rightarrow x_{i+1}$	Parallel (All tokens at the same time)	Speculative (Draft $\rightarrow$ Verify)
Context	Unidirectional	Bidirectional (Global context)	Bidirectional (Global in drafting)
Time Complexity	$O(N)$: Linear to sequence length	$O(K)$: Decoupled from sequence	Optimized via Rejection Sampling or draft
SER (Output coherency)	Strong	Moderate	Strong
Reversal Capability	Poor	Strong	Moderate to Strong

Table 1: Direct Comparison of the methods: Comparison of both autoregressive and diffusion based methods based on certain criteria.

time [2] [7]. Although one iteration of a diffusion model might take longer than a step of an autoregressive model, in theory, diffusion models scale better the longer the sequence is, as the length practically doesn't matter.

Another advantage that stems directly from the way DLMs generate text is that they have bidirectional context. This means that because multiple tokens are being generated independent of their order in the sequence, the model can "plan ahead" by basing earlier tokens on tokens that were generated much further into the sequence [13]. This advantage not only proves to be useful in creative writing, where the model should already plan towards the outcome of a narrative, but it also breaks the reversal curse that is present in ARMs. The reversal curse is the phenomenon that occurs if a model was trained on a sentence in the form "A is B", but won't generalize this knowledge to know that "B is A" [1]. Diffusion models such as LLaDA don't suffer from that reversal curse and even outperform models like GPT-4o in knowledge reversal [13].

Also building on the bidirectional context is an improved guidance of the generation. While autoregressive models have no prior knowledge of what will be placed further along the sequence, for a diffusion model, users can already define tokens in the middle of the output. This allows them to, for example, mention a certain structure in which the output should be generated Li et al. [7]. The diffusion model then takes this empty structure and fills in the missing tokens. The model could, for example, generate a poem where each verse starts with a certain letter Xiong et al. [17] or generate the output in a certain JSON structure, without having to fine tune and engineer the prompt. Additionally, DLMs do iterative refining, meaning they can go over the generated text multiple times and refine it [7]. This means they don't have to lock certain tokens in like ARMs but can accept tokens that have a high confidence and keep iterating over low-confidence tokens. This results in a low *token error rate* [2] which is commonly measured in *perplexity* [5]. Which means, that the model is more certain about each token locally, making it more fluent.

While being able to mimic ARMs, DLMs can be controlled in an easier manner. The style, sentiment, and toxicity can be influenced in a much easier way than in ARMs, the latter having to be fine-tuned to change the output. This changes the model's weights and deteriorates performance [12]. Furthermore, this fine-tuning process has to be done whenever a new application needs attributes to change, which would result in many dedicated models [12]. In contrast, DLMs can be controlled using methods presented by Lovelace et al. [12] or with techniques like CFG (classifier-free guidance), as stated in Li et al. [7]. This control over the language model might result in a more efficient and broader use of DLMs.

Finally, DLMs use the same framework as other modalities, like

image processing, incorporating multiple modalities into one model [7]. However, this aspect will not be further discussed in this report but can be observed in multimodal diffusion models like LLaDA-V You et al. [20].

3.2 The cost of diffusion

While the aforementioned parallelism of the token generation from DLMs is an advantage in performance, it also poses a problem. Generating, or in the case of MDMs unmasking tokens in parallel disregards the interdependencies between tokens as discussed by Feng et al. [2]. The model could end up generating all tokens at the same time, resulting in an incoherent mess of "thoughts" that don't quite fit together. This aspect is called SER, meaning *sequence error rate* and is measured by comparing the distribution of a generated sequence to the ground-truth distribution [2]. A poor SER results in the reasoning capabilities and logical consistency of the DLM output becoming useless. Recently, the need for a chain of thought generated by a language model is becoming increasingly important as it increases the abilities of language models significantly [15] [16]. This means that diffusion models won't be practical in the sense that they cannot be used for logical tasks like math or programming. At least not in this state, with the SER being that high.

The SER of the DLMs output can be reduced; however, this requires significantly more iterations that refine the generated sequence. While overall improving the output of the model, it should be noted, that the inference cost of a single iteration in a DLM is significantly higher than a step of an ARM. This results in an overall diminishment of the efficiency advantage, as shown by Feng et al. [2].

Another way of decreasing the SER, is to restrict the window of the generation for the model. Restricting the model to only generate subsequent blocks of tokens in parallel decreases the SER significantly. The DLMs become a middle ground between AR and diffusion by mimicking autoregressive models. Although Feng et al. [2] show that this does improve the output of the DLMs regarding TER and SER, they also show that the improvement isn't enough to surpass similarly sized AR models.

4 Overcoming the limitations

Solutions for the above-mentioned limitations of DLMs do exist and were published recently. There are different approaches that offer solutions to different problems, leveraging the diffusion models strengths differently.

4.1 Scaling and initializing the models

To be comparable to models with billions of parameters like LLaMA [4] or Qwen2.5 [14], Nie et al. [13] propose a Large Language Diffusion *Model* with mAsking, or LLaDA for short. This

Model	Size	Coding				Math	
		HumanEval	HumanEval+	MBPP	MBPP+	GSM8k	Minerva Math
LLaMA3.2	1B	17.68%	14.63%	26.60%	38.89 %	5.67%	1.92%
Qwen3	1.7B	48.17%	41.46%	55.80%	71.43 %	66.72%	29.74%
TiDAR	1.5B	43.29%	39.02%	41.40%	62.11 %	53.90%	25.48%
LLaMA3	8B	34.80%	-	48.80%	-	48.70%	16.00%
Qwen3	8B	64.63%	56.71%	69.40%	83.07%	81.80%	52.94%
LLaDA (Base)	8B	35.40%	-	40.00%	-	70.30%	31.40%
LLaDA (Reported)	8B	32.32%	27.44%	40.80%	51.85%	70.96%	27.30%
Dream	7B	57.90%	-	56.20%	-	77.20%	39.60%
TiDAR (Trust Diff)	8B	57.93%	55.49%	65.40%	80.95%	80.44%	51.64%

Table 2: Generative Evaluation Results: Comparison of the generation quality between ARMs and DLMs. Results are mainly taken from Liu et al. [9] with *LLaDA (Base)* and *LLaMA3 8B* being the results reported by Nie et al. [13]. The results from *Dream* are taken from Ye et al. [19].

Model	Size	ARC-e	ARC-c	Hellaswag	PIQA	Winograde
LLaMA3.2	1B	65.28%	36.35%	63.76%	74.43%	63.30%
Qwen3	1.7B	73.32%	44.62%	66.43%	75.63 %	64.96%
TiDAR	1.5B	77.78%	45.39%	65.26%	75.52 %	63.61%
LLaMA3	8B	-	53.10%	79.10%	80.60%	77.30%
Qwen3	8B	81.90%	53.16%	78.59%	79.22%	75.69%
LLaDA (Reported)	8B	73.78%	49.15%	71.05%	73.88%	74.66%
Dream	7B	83.90%	59.80%	73.30%	75.80%	74.50%
TiDAR (Trust Diff)	8B	84.18%	58.53%	76.36%	80.25%	76.48%

Table 3: Likelihood Evaluation Results: Comparison of common-sense reasoning between ARMs and DLMs. Results are taken from Liu et al. [9]. The results from *LLaMA3 8B* are taken from Nie et al. [13], while results from *Dream* are from Ye et al. [19].

masked diffusion model was trained from scratch and performs on par with LLaMA3 [4], Qwen2.5 [14] and more similarly sized models, as seen in Table 2. Furthermore, LLaDA excels at breaking the reversal curse, outperforming GPT-4o in reverse poem completion. However, it only outperforms them in reverse and not in forward poem completion. Even though it outperforms autoregressive models, it only does so in some tasks, which is being explained by Nie et al. [13] in the difference in training data. While LLaMA3 [4] was trained on 15T tokens, LLaDA was only trained on 2.3T. This is especially true for GPT-4o, which was trained on significantly larger datasets and more compute power. The point of LLaDA is to show, that scaling a masked diffusion model to a size of 8B parameters does improve the quality of the generation. Even though LLaDA does support flexible sampling, meaning it can in theory mimic autoregressive sampling, using the plain diffusion process yields the best performance and quality [13]. This proves that DLMs do offer scalability and perform better if they get trained more extensively. Additionally, techniques like classifier-free guidance (CFG) are also supported by LLaDA, giving users more control over the generation. Extending on LLaDA are models like Dream, which show even better improvement on almost all tests. It does this partially by

being initialized with the weights of an autoregressive model, namely Qwen2.5 7B [14]. This weight initialization for diffusion models with weights from ARMs was already done with LLaMA [4] and GPT, resulting in promising models like DiffuLLaMA and DiffuGPT, which perform on par with their autoregressive counterparts [3]. In addition, it is trained differently, as it uses a special noise schedule for training, using a novel approach called CART and not uniform distribution. This means that not all tokens are assigned the same noise level, or in other words, not every token has the same importance. Dream re-decides noise levels for every token based on their context information. For example, in the sentence "The apple is red!", the word "The" does not carry as much meaning as "apple" or "red", indicating that the models should not use too much time thinking about it. Lowering the noise level for these words will then result in the model using less steps to think about it. This training and weight initialization results in it needing fewer tokens, since the knowledge from Qwen2.5 [14] is already adopted. The results also show on par performance in comparison to models like LLaMA3 in generation, as shown in Table 2 and Table 3. The comparisons are done using different benchmarks ranging from coding, like HumanEval and

MBPP, over math like GSM8k and Minerva Math to common-sense reasoning, like ARC-e and Hellaswag. While LLaDA states that the model is in its current state slower than autoregressors as it doesn't benefit from optimizations like KV-Cache, Dream shows how they can match the accuracy of Qwen2.5 7B [14] with double the speed. Pure diffusion models like LLaDA and Dream enable further exploration of the diffusion paradigm as they establish a strong foundation for other research to build on. For example, Xiong et al. [17] use LLaDA-1.5 to demonstrate how diffusion models can be used for controlling the structure of the generated output of a model.

4.2 Hybrid approaches

Models like LLaDA and Dream are pure diffusion models at training and inference time. Although they offer promising results, another method of using the diffusion models is to combine ARMs and DLMs together.

4.2.1 Diffusion Guidance and STAR-LDM.

For example, Lovelace et al. [12] propose an architecture that attempts to bring the controllability of diffusion models to the established effectiveness of autoregressive models. They do this by using a diffusion sampler to generate a semantic proposal embedding, meaning a vector holding the meaning of a sequence, and using an ARM to effectively *elaborate* on that. This plays into the strengths of both model paradigms, as the DLM only generates the *behind-the-scenes feeling* of the text, while the ARM then uses this to generate the concrete text. Building on that, STAR-LDM uses a similar approach, first generating a *soft prompt* using a DLM. During the generation of the soft prompt, the ARM generation is paused, giving the diffusion model time to think [11], giving it the name Stop Think and Auto Regress. This approach effectively lets the model think and plan using diffusion but doesn't have it rely on the incoherent text generation of diffusion models. The focus of STAR-LDM is not primarily on improving the speed of the generation but on using the *soft prompt* of the diffusion model to control the generation. By letting the diffusion model guide the autoregressive model, text attributes like toxicity or sentiment can be controlled using lightweight classifiers. Using a plug-and-play functionality like that is only available for diffusion models, as autoregressive models must be retrained and fine-tuned [11]. STAR-LDM also enables the models to perform global semantic planning, increasing the likelihood of content-bearing words that capture the essence of the sentence semantics.

4.2.2 TiDAR.

While STAR-LDM and the model proposed in Lovelace et al. [12] effectively halt generation of text, the model proposed in Think in Diffusion Talk in Autoregression or TiDAR, integrates performance improvements into the hybrid architecture. TiDAR incorporates both an ARM and DLM into one big model, letting the DLM draft a sequence with the ARM using rejection sampling to selectively produce text. Liu et al. [9] states that both those processes are parallelized, leveraging the parallel generation power of GPUs to speed up generation compared to ARMs Qwen3 [18] and DLMs like LLaDA, Dream by 5.91x on average (in its 8B variant). This is done while also matching the generational quality of state-of-the-art models like Qwen3 [18] or LLaMA3.2 [4] as seen in Table 2 and Table 3. The key idea behind TiDAR is the already used technique of speculative decoding, which incorporates a smaller language model to draft a response Leviathan

et al. [6], which is then compared against what a larger model would answer. Speculative decoding is already established for autoregressive models. TiDAR leverages the technique to speed generation up using diffusion. They mention, that the main advantage of TiDAR is that both the DLM and ARM are in one model, giving it a high capacity to use the same weights and compute power. This allows TiDAR to draft the diffusion response in the same forward pass as the autoregressive rejection sampling. Another merit worth mentioning is that TiDAR can run in two different generation modes. These two modes differ in which model will be trusted more when selecting the tokens. TrustDiff means that the model trusts the diffusion prediction more, while TrustAR leads the model to trust the autoregressor over the diffusion model. Liu et al. [9] Show that for larger models, trusting the diffusion model more makes it perform better in planning-intensive tasks like math. This accentuates the ability of diffusion models to use long-term planning better than their autoregressive counterparts.

5 Discussion

Looking at the results of LLaDA and Dream compared to newer models like Qwen3 one can see how even newer diffusion models still lack in comparison to autoregressive models. Dream does improve on that but still doesn't beat Qwen3 in tasks like coding and math, as seen in Table 2 where the models were evaluated using widely known and accepted benchmarks like HumanEval, MBPP, GSM8K, and Minerva Math. Nie et al. [13] claim that the lack of performance of LLaDA and similar diffusion models is because of the lack of training data. While that may be true, they should also be compared to TiDAR which was only trained on 150B tokens [9] but performs roughly the same as Qwen3 and even surpasses it in some cases. Especially the comparison of common-sense reasoning shows how TiDAR outcompetes the pure diffusion models but also normal autoregressors in popular benchmarks, as shown in Table 3. This strongly suggests, that even though LLaDA and Dream offer a solid foundation and a proof of concept for diffusion models, one can still achieve better results with diffusion models using less training data.

Still, TiDAR doesn't leverage the controllability that diffusion models offer. LLaDA and Dream, however, offer CFG [13] and can inherently be guided by incomplete prompt data like an incomplete JSON object [17]. Additionally, STAR-LDM or the model proposed by Lovelace et al. [12] support control about toxicity or sentiment [11]. And although these models don't perform as well, they might still be suited for other applications, especially if pure diffusion models are adopting the weights of powerful autoregressive models [3]. Although this does require the existence of autoregressive models.

The improvement on knowledge reversal demonstrated by Nie et al. [13] is a strong argument for pursuing diffusion models, be it as a pure diffusion model with a large training set or in combination with autoregressive models. Knowledge reversal is not mentioned by Liu et al. [9], but LLaDA demonstrates the possibility of DLMs to generalize better than ARMs. If models like TiDAR also learn to reverse relations in the training data, hybrid models would have the definitive lead between pure diffusion models and hybrids.

6 Conclusion

Diffusion models offer many promising advantages over autoregressive models. Even though pure diffusion models have pitfalls

like their incoherent output and high SER, leading to generation times that are worse than those of autoregressive models, DLMs like Dream show that scaling and initializing the model weights using existing ARMs can still deliver strong results. However, where the paradigm of diffusion really shines is in applications like TIDAR, which use both DLMs and ARMs together, using both in their strongest department. This not only speeds up generation time but also improves coding, reasoning, and planning capabilities. Because of the recent findings that DLMs can be used to speed up generation and improve on the capabilities of ARMs, they may establish themselves as a contender or extension to the paradigm of autoregressive models. This is especially true if the models are used according to their strengths, meaning that while models that specialize in the speed aspect of diffusion might be good for chat applications, diffusion models that generalize better could be used in big data analysis and reproduction.

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